

Public Economics (ECON 131)

Final Exam Review

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1 True/False Statements

1. It would be foolish for the United States to introduce a wealth tax on the rich because the European experience has shown that such taxes are easy to avoid or evade.
2. Goods whose demand is price inelastic should be taxed more.

2 Labor Income Taxation + Labor Income Responses to Taxation

2.1 Yagan Midterm 2 Spring 2026, Question #3

A city has 1,000 working-age residents. Each resident decides whether to work or not. A worker earns pre-tax labor income $z > 0$; a non-worker earns $z = 0$. Initially there are **no taxes or transfers**.

(a) The city introduces a **Negative Income Tax (NIT)**:

- Every resident receives a **guaranteed income of \$20** (even with zero earnings).
- The guarantee is **phased out at 50 cents per dollar** of earnings.
- There are **no further taxes** above the point where the guarantee is fully phased out.

At what level of earnings z does the guarantee fully phase out? Write after-tax-and-transfer income $c(z) = z - T(z)$ for each segment, and draw the budget constraint with pre-tax earnings z on the x-axis and c on the y-axis. Label the y-intercept, any kink points (with coordinates), and the slope of each segment. (12 points)

- (b) Using the NIT from part (a), compute the **participation tax rate**

$$\tau_p = \frac{T(z) - T(0)}{z}$$

for a worker earning $z = 20$ and for a worker earning $z = 80$. Which faces the higher participation tax rate? In one sentence, explain intuitively why a system with a phased-out guaranteed income creates especially high participation tax rates for low earners. (12 points)

- (c) Before the NIT was introduced, there were 700 workers and 300 non-workers. After the NIT is introduced:

- 100 workers who had been earning $z = 20$ each **stop working**, discouraged by the high participation tax rate.
- 500 workers who earn $z = 80$ each continue working.
- The remaining 100 workers who earn $z = 20$ each also continue working.
- The 300 original non-workers remain out of work.

Compute the government's **total cost** of the NIT (total grants paid out, net of any phase-out clawback). (12 points)

(d) The city replaces the NIT with an **Earned Income Tax Credit (EITC)**:

- **No payment to non-workers:** $c(0) = 0$.
- **Phase-in** ($z = 0$ to $\$20$): workers receive a subsidy of $\$0.50$ per dollar earned.
- **Plateau** ($z = \$20$ to $\$40$): workers receive a flat credit of $\$10$.
- **Phase-out** (z above $\$40$): the credit is reduced by $\$0.25$ per dollar of earnings above $\$40$.
- No other taxes.

At what earnings level is the credit fully phased out? Draw the budget constraint (same axes as part (a)). Label the y-intercept, all kink points (with coordinates), and the slope of each segment. (12 points)

(e) Motivated by the EITC structure in part (d), the government considers a modest reform to the NIT from part (c): it **reduces the phase-out rate from 50% to 25%** on the first \$20 of earnings. Above \$20, the phase-out rate remains 50%. All else stays the same (the guaranteed income is still \$20). (12 points total)

(i) What is the participation tax rate at $z = 20$ under this reformed NIT? Compare it to the original NIT participation tax rate from part (b). (4 points)

(ii) The reform has a **mechanical cost**: the 100 workers at $z = 20$ who were already working now receive a larger net transfer. How much extra does each one cost the government? What is the total mechanical cost? (4 points)

(iii) Suppose that among the 200 residents with potential earnings $z = 20$, the **extensive-margin elasticity of the number who work** with respect to the net-of-tax rate $1 - \tau_p$ is

$$\varepsilon = \frac{\% \Delta \text{workers}}{\% \Delta (1 - \tau_p)} = 2.$$

Using the original NIT in part (c) as the baseline, compute how many of the 100 workers who had dropped out would re-enter when the participation tax rate falls from 50% to 25%. Then compute the **fiscal saving** from each returning worker: compare what they cost as a non-worker versus what they cost as a worker under the reformed NIT. What is the total extensive-margin saving? Does it cover the mechanical cost from part (ii)? Explain in 1–2 sentences why this type of reform can be a “free lunch.” (4 points)

2.2 Saez Midterm Spring 2025, Question #3

Assume that all individuals have the same utility function over consumption and **labor** given by:

$$U(c, l) = c + \theta \ln(16 - l)$$

where c represents consumption, l represents hours of labor, and θ is a constant parameter that reflects an individual's distaste for labor hours. Also, $\ln(\cdot)$ denotes the natural logarithm (this can also be denoted by $\log(\cdot)$). Suppose the only income that individuals have is from labor income, and that they work at an hourly wage w which is taxed at rate t .

- (a) Assuming the government imposes a tax on labor income at rate t , solve for the individual's optimal labor supply as a function of w , θ , and t . (2 Points)

- (b) For this part (b) only, assume the following:

- There are 100 individuals in this society with the utility function given above.
- All individuals have $\theta = 24$.
- There are 50 low-wage individuals and 50 high-wage individuals. The low-wage individuals earn wage $w = 4$, while the high-wage individuals earn wage $w = 8$.

Calculate how much revenue the government will raise if it imposes a 25% labor income tax ($t = 0.25$) on all 100 individuals. (2 Points)

- (c) The government creates a universal basic income (UBI) program to support individuals with low or zero labor income, where the UBI phases out after a certain threshold of income. The program is financed by imposing a labor tax on higher earners.

More specifically:

- The UBI program provides all individuals a lump-sum grant of \$20.
- The grant does not phase out for the first \$24 of labor income.
- After the first \$24 of labor income, the grant is phased out at a 50% rate.
- After the grant phases out entirely, labor income from that point up to \$120 is taxed at a 25% rate.
- After \$120, any additional labor income is taxed at a 50% rate.

Graph the budget constraint (*Hint: there are four brackets*). Put pre-tax income ($Z = wl$) on the x-axis and after-tax income $c = Z - T(Z)$ on the y-axis. **Label the x and y values of each kink point, and the slope of each of the four segments of the graph.** (2 Points)

- (d) For each of the four brackets in your budget constraint, indicate the sign (direction) of the substitution and income effects on an individual's choice of labor supply. Also indicate the sign (direction) of the total (combined) effect of the two, if it is possible to know for certain, or indicate with a question mark if the total effect is uncertain. (2 Points)

	Bracket 1	Bracket 2	Bracket 3	Bracket 4
Substitution Effect:				
Income Effect:				
Total Effect:				

- (e) A new administration is elected and they change the UBI policy:

- The grant is cut from \$20 to \$8.
- Workers are subsidized on the first \$24 of earnings with a new credit that pays \$0.50 per dollar of pre-tax labor income. This new credit phases out at a 50% rate after the first \$24 of income.
- All else stays the same.

Theoretically, how do you expect this policy change (i.e. moving from the old UBI policy to this new policy) to affect the extensive and intensive labor supply choices of individuals at and near the bottom of the income distribution? (2 Points)

Keep answers succinct. No need for any calculations.

3 Local Public Goods

3.1 Saez Final Exam Spring 2023, Question #2

The city of Berkeley has decided to build a new BART station near the Greek Theater to improve access to public transportation on the east side of the university campus. The city of Berkeley plans to fund the functioning of this new BART station, and notably improves the frequency of the BART trains going through the station, from the contributions of only two individuals: a UCB student (S) and a UCB professor (P). Each of the two individuals has a utility function over private goods (x_i) and total number of BART trains going through the new station everyday (B), of the form:

$$U_i(x_i, B) = (x_i - 3)(B - 4) \quad (1)$$

The total number of BART trains going through the station everyday B , is the sum of the number of trains paid for by each individual, the student and the professor: $B = b_s + b_p$. The student has an income of \$5 and the professor has an income of \$11. Both the private good and a BART train have a price of \$1.

- (a) How many BART trains will serve the new BART station everyday if the city of Berkeley does not intervene? How many are paid for by the student? By the professor? (4 points)

(b) What is the socially optimal number of BART trains? (4 points)

(c) Why do we observe a difference between (a) and (b) ? (2 points)

- (d) Suppose the city of Berkeley is not happy with the private equilibrium and decides to provide an additional BART train in addition to what the student and the professor choose to provide on their own. This extra train is paid for by imposing a \$1 lump-sum tax on the professor only. What is the new total number of trains? How does your answer compare to (a)? Did the city of Berkeley achieve the social optimum with this plan? Why or why not? (4 points)
- (e) Propose instead a mechanism the city of Berkeley could use to achieve the social optimum. (4 points)
- (f) Does the BART system fit the characteristics of a pure public good? Why or why not? (2 points)